

The graph of an exponential function is shown.

1. Find the value of a by substituting the point $(0, 2)$ in for x and y in $y = a(b)^x$.

$$[2] = a(b)^{[0]}$$

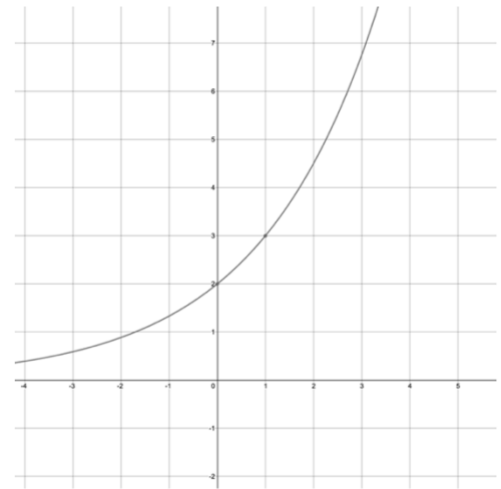
$$a = 2$$

2. Use the value of a found in question 1 and the point $(1, 3)$ to determine the value of b .

$$y = 2(b)^x$$

$$3 = 2(b)^1$$

$$b = 1.5$$



3. Write the function for the graph in both forms.

$f(x) = a(b)^x$	$f(x) = a(1 \pm r)^x$
$f(x) = 2(1.5)^x$	$f(x) = 2(1 + 0.5)^x$

The table of an exponential function is shown.

x	0	1	2	3	4
$f(x)$	-7	-14	-28	-56	-112

4. What is the initial value, a ?

$$-7 = a(b)^0$$

$$a = -7$$

5. How does the value of $f(x)$ change as the x -value increases by 1?

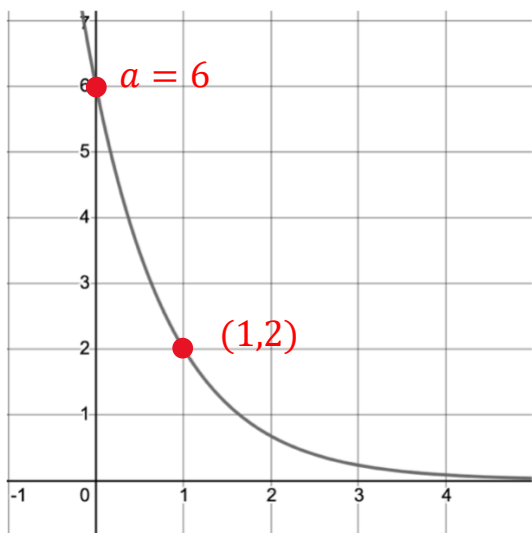
As the x -value increases by 1, the y -value or $f(x)$ value is multiplied by a factor of 2.

6. Write the function in the form $f(x) = a(b)^x$

$$f(x) = -7(2)^x$$

For questions 7 – 10, write an equation from the table or graph by finding the initial value, a , and common ratio, b .

7.



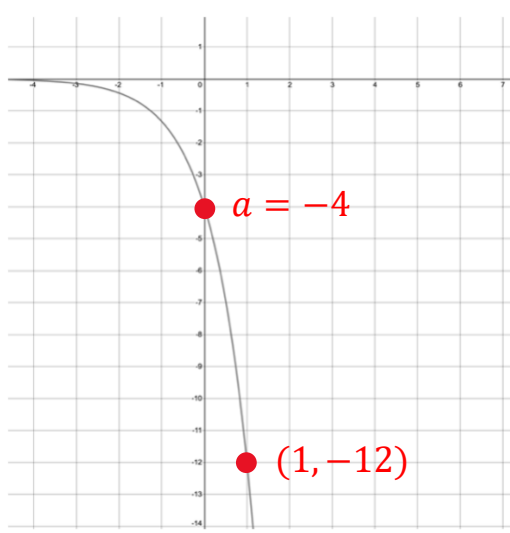
Function

$$2 = 6(b)^1$$

$$b = \frac{1}{3}$$

$$f(x) = 6\left(\frac{1}{3}\right)^x$$

8.



Function

$$-12 = -4(b)^1$$

$$b = 3$$

$$f(x) = -4(3)^x$$

9.

x	y
0	-5
1	$-\frac{25}{4}$
2	$-\frac{125}{16}$
3	$-\frac{625}{64}$
4	$-\frac{3125}{256}$

$a = -5$

$$\times \frac{5}{4}$$

$$\times \frac{5}{4}$$

$$\times \frac{5}{4}$$

$$\times \frac{5}{4}$$

Function

$$\frac{-\frac{25}{4}}{-5} = \frac{-5(b)^1}{-5}$$

$$b = \frac{5}{4} \text{ or } 1.25$$

$$f(x) = -5\left(\frac{5}{4}\right)^x \text{ or}$$

$$f(x) = -5(1.25)^x$$

10.

x	y
0	3
1	2.1
2	1.47
3	1.029
4	0.7203

$a = 3$

$$\times 0.7$$

$$\times 0.7$$

$$\times 0.7$$

$$\times 0.7$$

$$\times 0.7$$

Function

$$\frac{2.1}{3} = \frac{3(b)^1}{3}$$

$$b = 0.7$$

$$f(x) = 3(0.7)^x$$